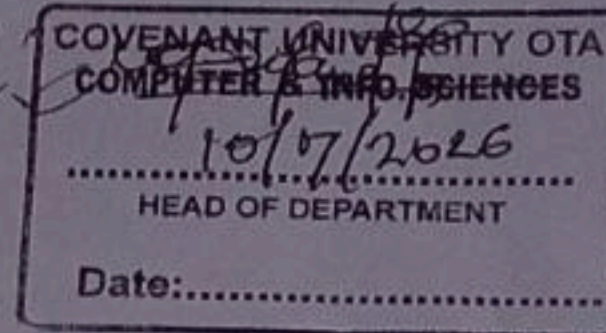




COVENANT UNIVERSITY
CANANLAND, KM 10, IDIROKO ROAD,
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TITLE OF EXAMINATION:

COLLEGE: Science and Technology

DEPARTMENT: Computer & Info. Sciences

SESSION: 2025/2026

COURSE CODE: CSC241

COURSE TITLE: Python I

Instruction: Answer any Four (4) Questions.

B.Sc. EXAMINATION

SEMESTER: OMEGA

CREDIT UNIT: 3

Time Allowed: 3 Hours

Question One

- Briefly describe how the *and* operator works in Python (use examples) [2.5 marks]
- What is a string in Python? List and briefly explain four common string methods used in Python. Provide an example for each method. [5 marks]
- Identify the Errors in the following code segment [3 marks]

```
1 person = { "name": 'John Doe', 'age': 30, 'city':  
2     'New York', 'Email': 'john.doe@example.com' }  
3  
4 print("Name:", person['Name'])  
5 print("Age:", person['Age'])  
6 print("City:", person['City'])  
7 print("Email:", person['email'])  
8  
9 person['Age'] = "35"  
10 person['phone'] = '123-456-7890'  
11 del person['email']  
12  
13 print("Updated Person Dictionary:")  
14 print(person)  
15
```

- Write a Python program that implements a mathematical model to calculate Body Mass Index (BMI). Use the formula: $BMI = W / H^2$, where: W = weight in kilograms, H = height in meters. [3.5 marks]

Requirements:

- Classify the BMI:
 - $BMI < 18.5 \rightarrow$ Underweight
 - $18.5 \leq BMI < 25 \rightarrow$ Normal weight

- $25 \leq \text{BMI} < 30 \rightarrow \text{Overweight}$
- $\text{BMI} \geq 30 \rightarrow \text{Obese}$
- 2. Display the BMI value with the BMI class.
- e) Write a Python program that calculates the future value of an investment using compound interest. The program should display A and the interest earned (A - P) to two decimal places. [3.5 marks]

Use the formula: $A = P(1 + r/n)^{nt}$ where:

- P = principal
- r = annual interest rate (decimal)
- n = number of times interest is compounded per year
- t = time in years

Question Two

- a) What is a count-controlled loop in Python? [3.5 mark]
- b) Explain the differences among the main types of control structures in Python. Give one example of each. [4.5 marks]
- c) Identify the Errors in this code segment and write the corrected version. [3 marks]

```

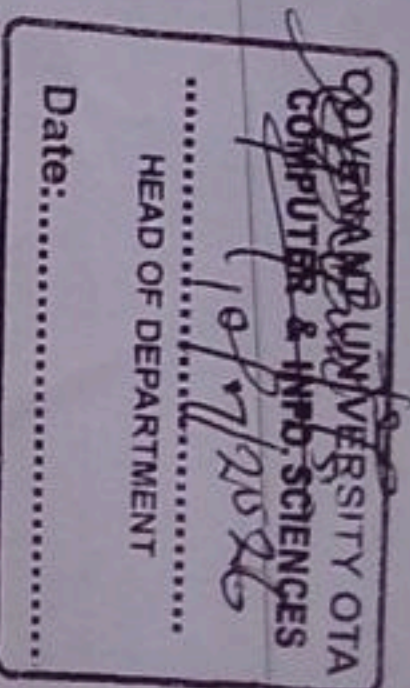
1 my_string = "Serve Jesus Now"
2 print("Original String:", mystring)
3
4 first_character = my_string[1]
5 last_character = my_string[-0]
6 print("First Character:", first_character)
7 print("Last Character:" last_character)
8
9 substring = my_string[17:]
10 print("Substring:", substring)
11
12 greeting = "Hi"
13 name = "Alice"
14 message = greeting + ", " + name + "!"
15 print("Message:", message)
16
17 length = len(mystring)
18 print("Length of the String:", length)
19
20 count_e = my_string.count('e')
21 print("Count of 'e':", count_e)
22

```

- d) A secondary school is developing a simple Python program to assist teachers in managing students' assessment activities. The school has provided the following grading scheme: [6.5 marks]

- 70 - 100: Grade A
- 60 - 69: Grade B
- 50 - 59: Grade C

- 45 - 49: Grade D
- 0 - 44: Grade F



Given that the correct password for accessing the system is: admin123, write a Python program that uses appropriate control structures to perform the required tasks.

Requirements:

- i. Prompt the user to enter a password and continue requesting the password until the correct password (admin123) is entered.
- ii. Display "Access Denied" for every incorrect password entered and "Access Granted" when the correct password is supplied.
- iii. Prompt the user to enter a student's examination score.
- iv. Validate that the score is between 0 and 100. Display an appropriate error message if the score is outside this range.
- v. If the score is valid, determine and display the student's grade using the given grading scheme.
- vi. Prompt the user to enter a positive integer and use a for loop to display its multiplication table from 1 to 12 in a neatly formatted manner.
- vii. Using a while loop, repeatedly prompt the user to enter numbers until 0 is entered.
- viii. After the user enters 0, display: The total sum of all numbers entered (excluding 0), and The total number of valid numbers entered.

Question Three

- a) Mention five (5) benefits of modularizing a program with functions [2.5 marks]
- b) What is a function in Python? Why are functions important in programming? Differentiate between built-in functions and user-defined functions. Distinguish between a Void and Value-returning function in Python (with examples) [6 marks]
- c) What will list1 and list2 contain after the following statements are executed? [3 marks]
 - i. list1 = [1, 2] * 2
 - ii. list2 = [3]
 - iii. list2 += list1
- d) The Computer Science Department organized a beginner Python Programming Workshop for undergraduate students. You have been asked to write a simple Python program to manage the workshop information. The organizers provided the following records:
 - Registered Students: Esther, John, Paula, Paul, Peter, Ada, John, Mary
 - Workshop Details: "Python Programming Workshop", "Room 204", "Monday", 30
 - Students Who Attended: Esther, John, Mercy, Paul, Esther, John, Paula
 - Student Scores:
 - Esther scored 60
 - John scored 55
 - Mercy scored 65
 - Paul scored 80

Write a Python program that uses the most suitable Python collection for each type of data.

Requirements:

[6 marks]

- i. Store the registered students using the best collection.
- ii. Store the workshop details using the best collection.
- iii. Remove duplicate names from the attendance record.
- iv. Store each student's score using the best collection.
- v. Display the workshop details.
- vi. Display the unique list of registered students.
- vii. Display the unique list of students who attended.
- viii. Display students who registered but did not attend.
- ix. Display students who attended but were not registered.
- x. Display students who scored 60 and above.

[5 marks]

Question Four

- a) Strings are *Immutable*. Explain and with reasons.
- b) A program contains the following function definition:

```

2
3 - def cube(num):
4     return num * num * num
5

```

- i. Write a statement that passes the value 4 to this function and assigns its return value to the variable result. [1 mark]
- ii. What will be the value in result [1 mark]

- c) A secondary school has decided to computerize the management of students' academic records. You have been asked to develop a simple Python program that uses an SQLite database to store, retrieve, update, and manage students' records. The school requires that all records be stored in a database named school.db in a table named student. The table should contain the following fields: StudentID (INTEGER), StudentName (TEXT), Class (TEXT), and Score (REAL). The school has provided the following student records: [10.5 marks]

StudentID	StudentName	Class	Score
101	David James	SS1	78
102	Mary Johnson	SS2	85
103	Peter Okoro	SS1	67
104	Grace Bello	SS3	92
105	Esther Ade	SS2	74

Write a Python program that uses SQLite to manage the students' records.

Date:.....

Requirements:

- i. Create an SQLite database named `student` and do not already exist.
- ii. Insert the given student records into the student table.
- iii. Retrieve and display all student records.
- iv. Retrieve and display only the records of students in SS2.
- v. Retrieve and display the following statistics: Total number of students, Highest score, and Average score.
- vi. Update the score of Peter Okoro from 67 to 75.
- vii. Delete the record of the student whose StudentID is 105.

Question Five

- [6 marks]
- [2 marks]
- a) What is file handling in Python?
- b) List and explain four (4) file access modes used in the open() function.
- c) Mention the difference between reading from a file and appending to a file.
- d) The Computer Science Department conducted a Continuous Assessment (CA) for CSC241 students. The department requires that all student records be stored in a single text file throughout the program. The file created at the beginning of the program must be used for all subsequent operations. The assessment records should contain the following information for each student: Name, Matric Number, Department, and Score. The records should be stored in a text file named: *your_matic_number_CSC241.txt*. (Replace your_matic_number with your actual matric number. For example: 23CG012345_CSC241.txt.) Write a Python program that uses file handling operations to manage the students' assessment records. A sample record is shown below:

- Name: John Doe, Matric No: 23CG012345, Department: Computer Science, Score: 78
Name: Mary James, Matric No: 23CG012346, Department: MIS, Score: 85

Requirements:

- i. Prompt the user to enter the records of 10 students and save them in the specified text file.
- ii. Using the same file created in (i), prompt the user to enter the records of 5 additional students and append the new records to the file without deleting or modifying the existing records.
- iii. Using the same file, read and display all student records stored in the file.
- iv. Using the same file, read and display only the first five student records stored in the file.

Question Six

- a) Given the following code segment

```

1 2 ~ nested_list = [
2     [1, 2, 3],
3     [4, 5, 6],
4     [7, 8, 9],
5     [10, 11, 12, 13],
6     [14, [15, 16, 17], 18],
7     [19, [20, [21, 22], 23], 24]
8 ]
9
10

```


- i. What is the value at index [3][2]?
- ii. `print (len (nested_list[4]))`
- iii. What is the value at index [5][1][1][0]?

- b) A fitness club organized a Weekly Step Challenge to encourage members to maintain an active lifestyle. You have been asked to develop a Python program that tracks a participant's daily step count over a specified number of days and determines whether the participant achieved the overall fitness milestone. The club requires that the final performance report be displayed using a void function named `display_achievement_summary()`, which receives the participant's name, total steps taken, and overall target as arguments. Write a Python program that uses a void function to evaluate and display the participant's fitness achievement.

[7 marks]

Requirements:

- i. Prompt the user to enter their name, the number of days to track, and the daily step goal.
- ii. Use a loop to prompt the user to enter the number of steps taken for each day.
- iii. Calculate the total number of steps taken during the tracking period.
- iv. Calculate the overall target by multiplying the daily step goal by the number of days.
- v. Create a void function named `display_achievement_summary(user_name, actual_total, target_total)`.
- vi. Pass the participant's name, total steps, and overall target to the function.
- vii. Inside the function, display: the participant's name, the total number of steps taken, and a message indicating whether the participant achieved or did not achieve the overall fitness milestone.

Sample Interaction

```
Enter your name: Alice
Enter the number of days to track: 3
Enter your daily step goal: 10000
```

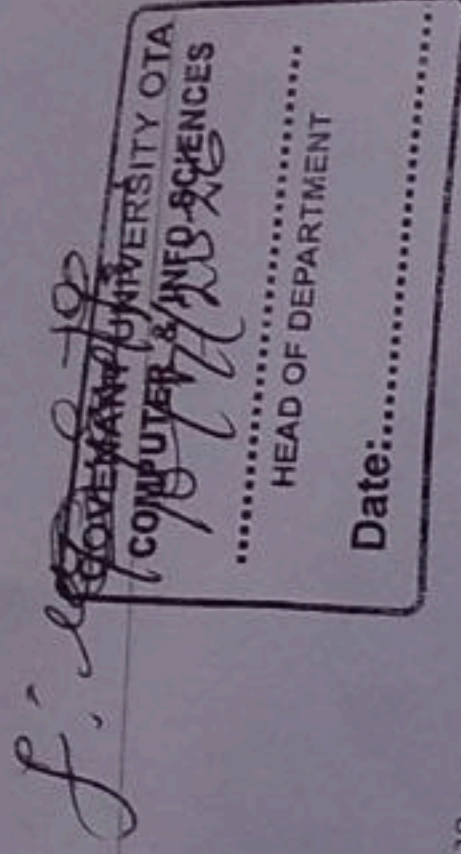
```
Enter steps for day 1: 12000
Enter steps for day 2: 9000
Enter steps for day 3: 11000
```

```
--- Progress Report for Alice ---
Total Steps: 32000
Milestone Achieved!
```

- c) A commercial renovation company specializes in installing flooring in circular meeting rooms for corporate clients. You have been asked to develop a Python program that estimates the total cost of a flooring project. To promote code reusability and modular programming, the company requires that the program be developed using value-returning functions. The area of the meeting room is calculated using the formula: $\text{Area} = \pi r^2$ where π should be obtained from the `math` module. The total project cost is calculated as: $\text{Total Cost} = (\text{Area} \times \text{Flooring Price}) + \text{Labor Fee}$. Write a Python program that uses value-returning functions to estimate the cost of the flooring project.

Requirements:

[7 marks]



f
Date: 10/17/2026

- that prompts the user to enter:
- Create a value-returning function named `get_data()` that prompts the user to enter:
Room radius, Flooring price per square meter, and Labor fee.
Return the three input values from `get_data()` to the main program.
 - Create a value-returning function named `calculate_area(radius)` that calculates and returns the area of the circular room using `math.pi`.
 - Create a value-returning function named `calculate_total(area, material_price, labor_fee)` that calculates and returns the total project cost.
 - In the main program, call the three functions appropriately to obtain the required values and calculate the final project estimate.
 - Display the total calculated project cost.
 - If the total project cost exceeds \$10,000, display the status message: Requires Senior Management Approval. Otherwise, display an appropriate message indicating that Senior Management Approval is not required.

Sample Interaction

Enter room radius (meters): 12
Enter flooring price per square meter: 20
Enter labor fee: 1500

--- Project Estimate ---
Total Calculated Cost: 10546.90
Status: Requires Senior Management Approval.